1-Week-Ahead Weekly Closing Price Prediction

Bai Yu · Team project

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TEAM PROJECT

Simple flowchart

Fetch Data
V
Cleaning & Feature Engineering
V
Train/Test Split
V
Models
V
Evaluation

Data: 2019–2025 from Yahoo Finance/FRED

```
# Yahoo Finance data source
TICKERS_YF = {
    PRIMARY_KEY: "JPM",              # JPMorgan Chase
    "bac": "BAC",                    # Bank of America
    "c": "C",                        # Citigroup
    "wfc": "WFC",                    # Wells Fargo
    "xlf": "XLF",                    # Financial sector ETF
    "gold": "GC=F",                  # Gold futures
    "ftse": "^FTSE",                 # FTSE 100 (UK)
}
```

1 Fetch Data

Data from Yahoo Finance and FRED



2019-2025

70k+

~

700

Yahoo Finance
and
Federal Reserve Economic Data

2 Cleaning data

We performed systematic data cleaning, which included:

01

Structural conversion

Fixed data formats, converted columns to the correct data types, and removed any structural errors.

02

Filling missing values

Filled missing values using forward-fill, time-based interpolation, and median as a fallback (especially for market holidays).

03

Outlier capping

We capped extreme values to reduce the influence of outliers while keeping all data points.

04

Removing invalid values

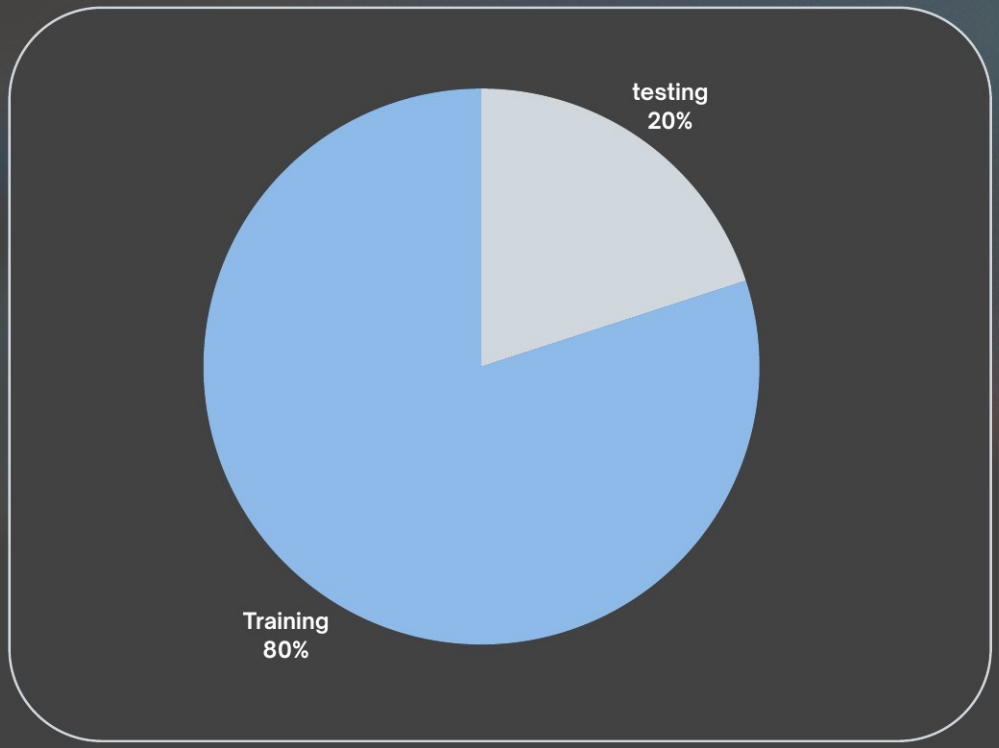
Removed duplicate dates and handled invalid values such as infinity (inf) and NaN.

3 Train/Test Split

We used a strict time-series split:

- 80% for training (2019–2024)
- 20% for testing (2024–2025) to avoid any lookahead bias.


```
Dataset ready: {
  "n_train": 569,
  "n_test": 143,
  "n_features": 15,
  "train_index_start": "2019-03-05",
  "train_index_end": "2024-08-13",
  "test_index_start": "2024-08-19",
  "test_index_end": "2025-12-29",
```

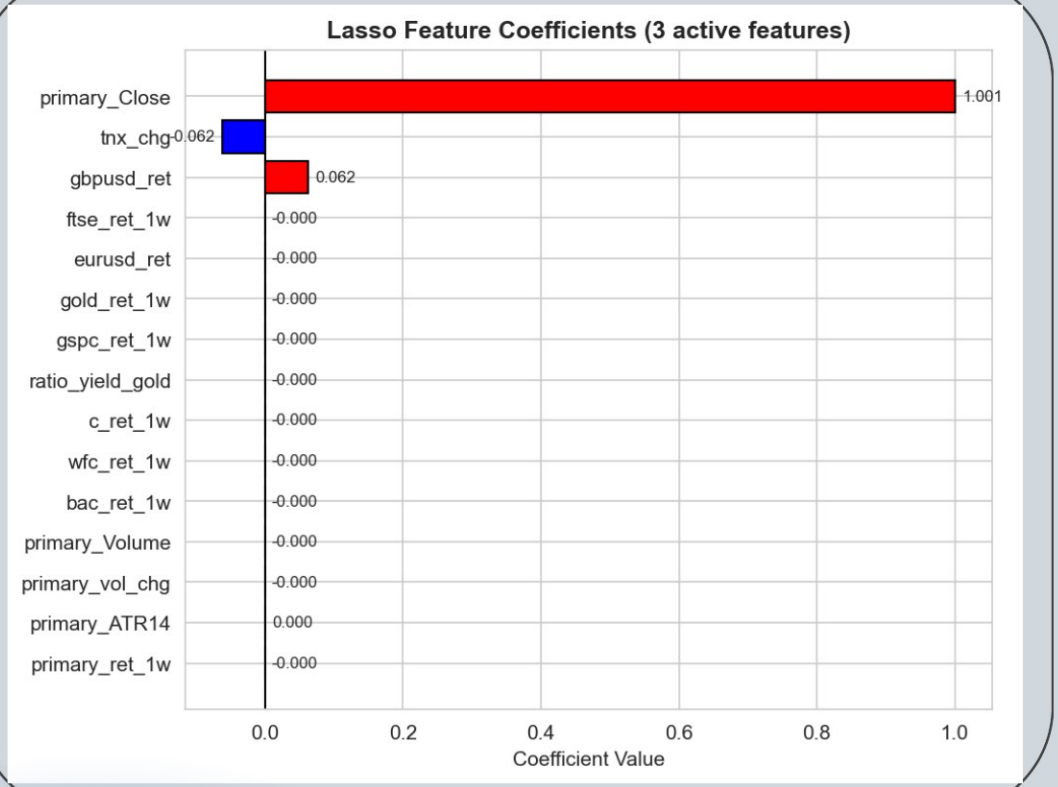
4 Models

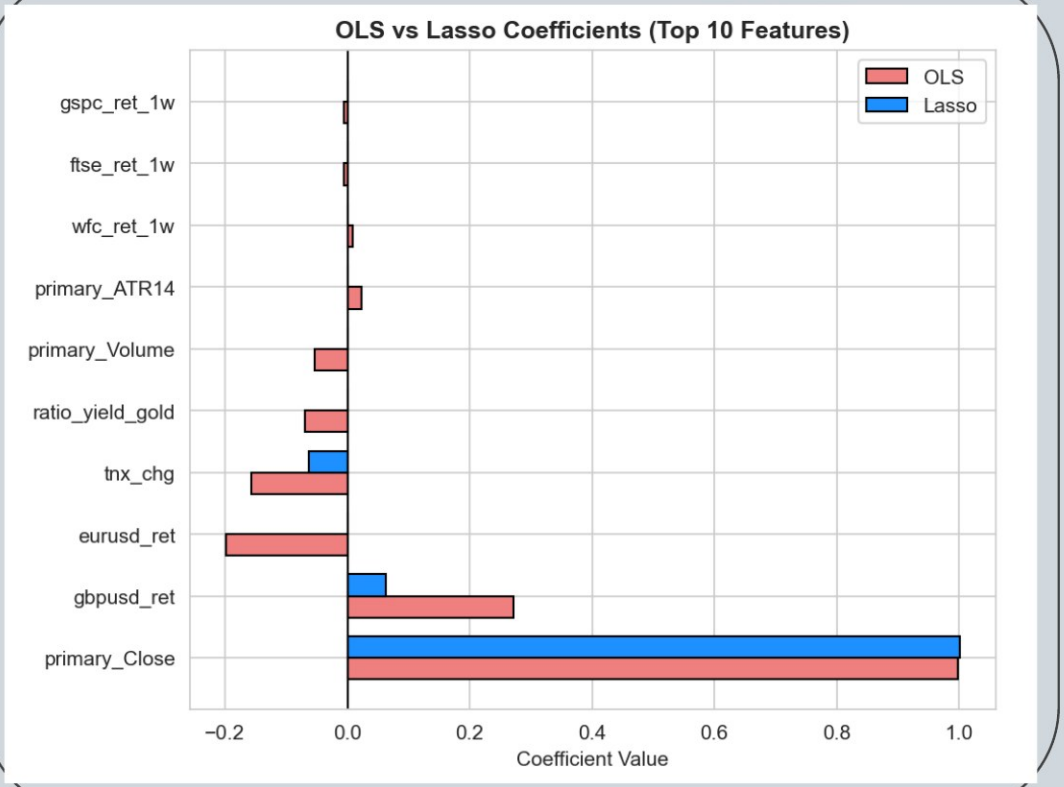
Presented by :
Bai Yu

Six different model

Model	RMSE	MAE	R^2	vs NPB RMSE
~~NPB (Baseline)~~	~~4.7677~~	~~2.4963~~	~~0.9941~~	—
OLS	4.7620	2.5188	0.9941	+0.11%
Ridge	4.7614	2.5168	0.9941	+0.13%
Lasso	**4.7537**	**2.5487**	**0.9942**	**+0.29%**
Random Forest	4.7620	2.5780	0.9941	+0.12%
XGBoost	4.8985	2.8003	0.9938	-2.74%

Lasso Model vs OLS





Lasso use L1 Regularization based on OLS

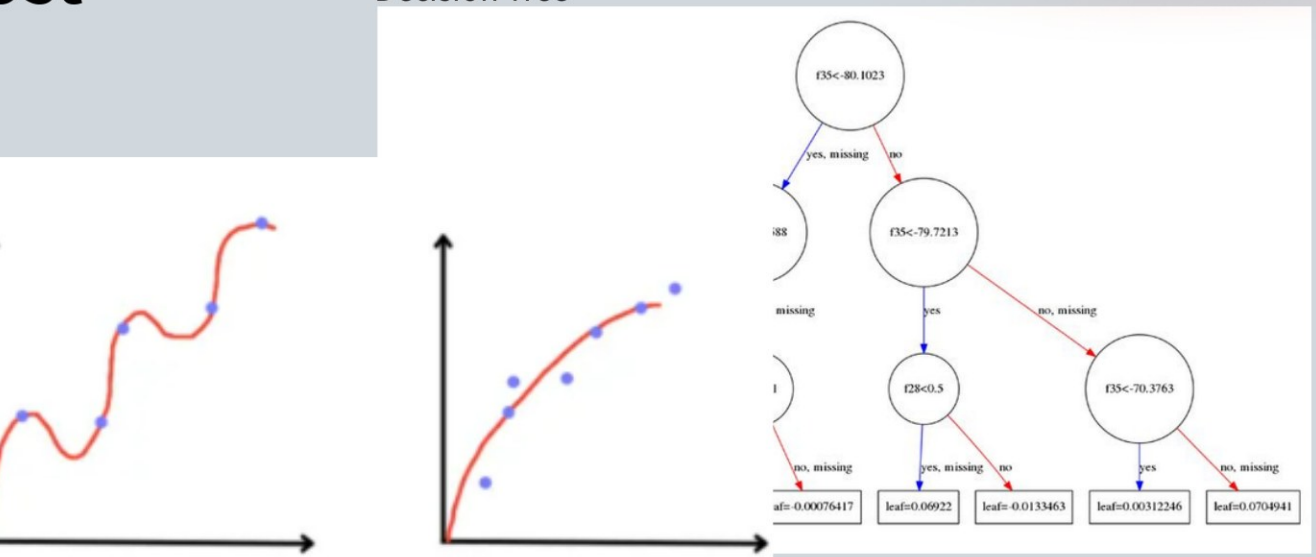
- **Low-contribution features: reach exactly 0**
- **High-contribution features: are barely pushed to 0**

Lasso Model vs XGBoost

XGBoost: Very High St
Complexity

- **Tree Ensemble Model Co**
  many decision trees (e
  trees) working together.
- **Complex Tree Structu**
  tree has deep and com
  splits to capture
  patterns.
- **More Complex Training**
  Trees are built sequen
  each new tree tries to correct the errors of all previous trees.

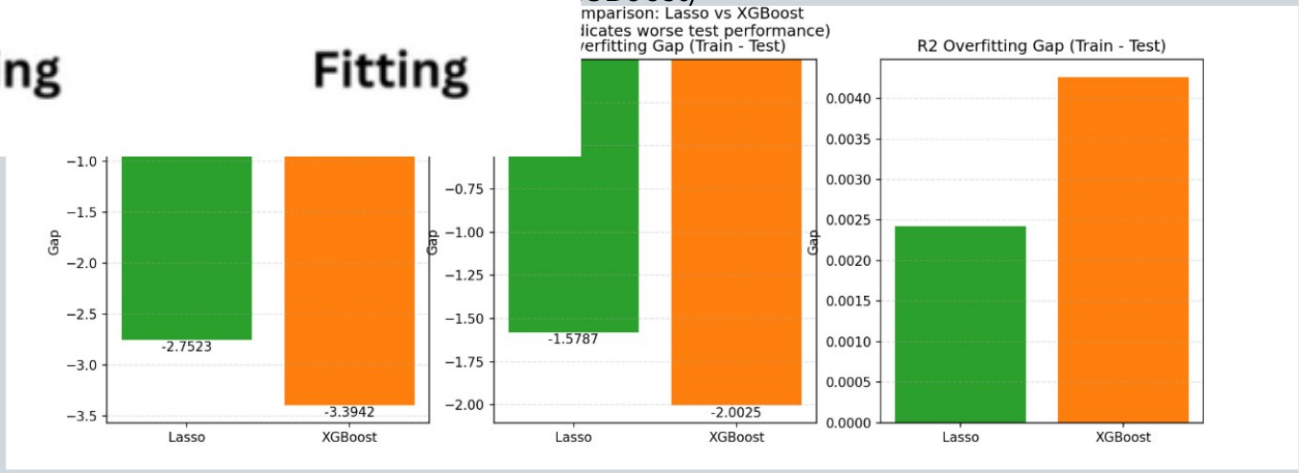
Decision Tree



Overfitting

Fitting

GBoost)



Simple flowchart - 5Evaluation

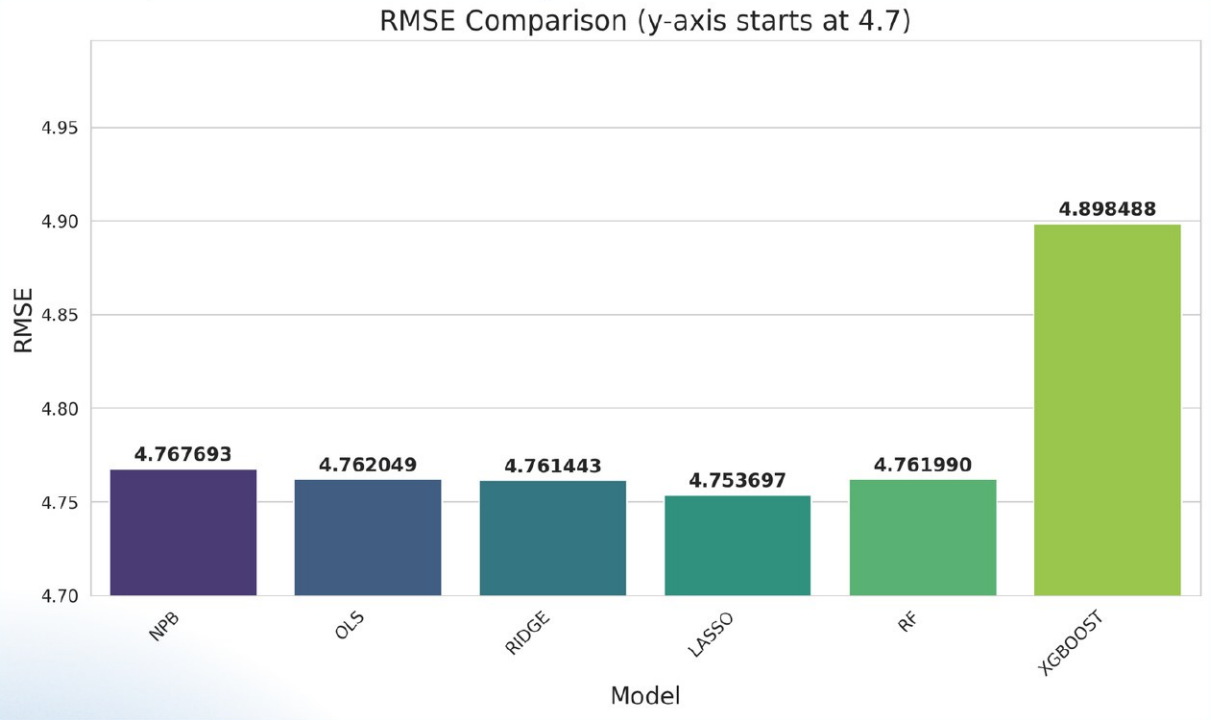
TEAM PROJECT

RMSE (Root Mean Squared Error)

Formula:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

RMSE (Root Mean Squared Error)



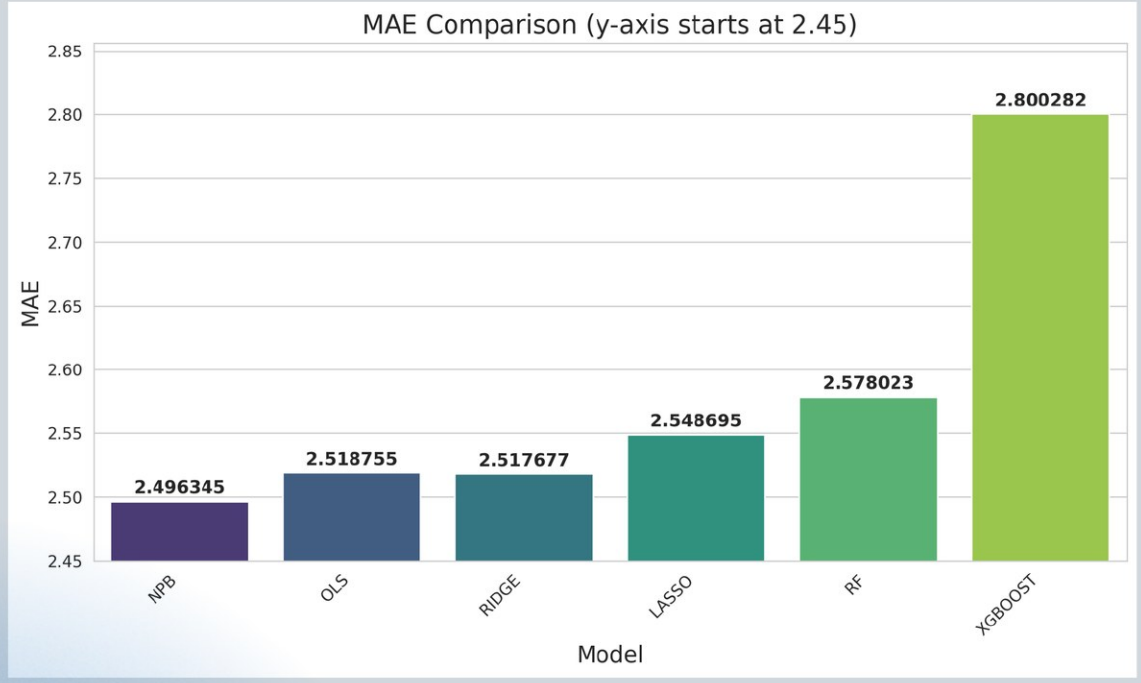
Lower RMSE = better !

Higher RMSE = worse !

- **Lasso: RMSE = 4.754** ⭐
- **XGBoost performed the worst: RMSE = 4.898** ❌

5 Evaluation

MAE (Mean Absolute Error)



MAE (Mean Absolute Error)

Formula:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i|$$

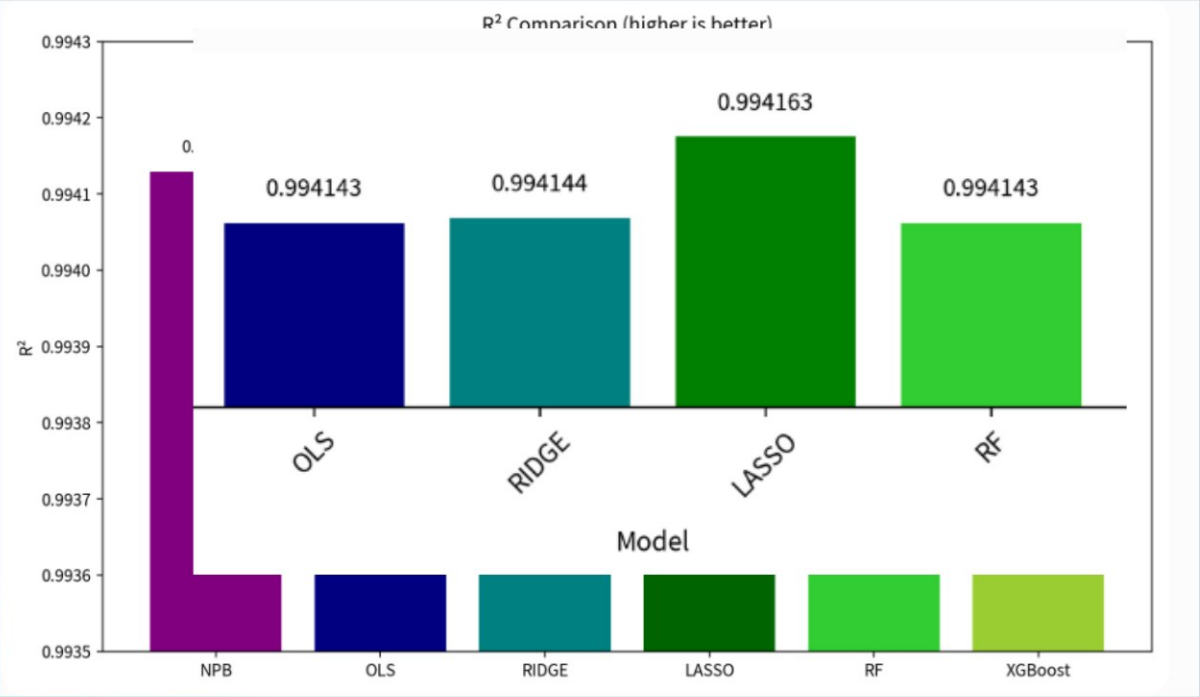
Lower MAE = better !

Higher MAE = worse !

- **Ridge : MAE = 2.496** ⭐
- **Lasso: MAE = 2.549** ⭐
- **XGBoost performed the worst: MAE = 2.800** ❌

Simple flowchart - 5Evaluation

R^2 (R-squared)



R^2 (R-squared)

Formula:

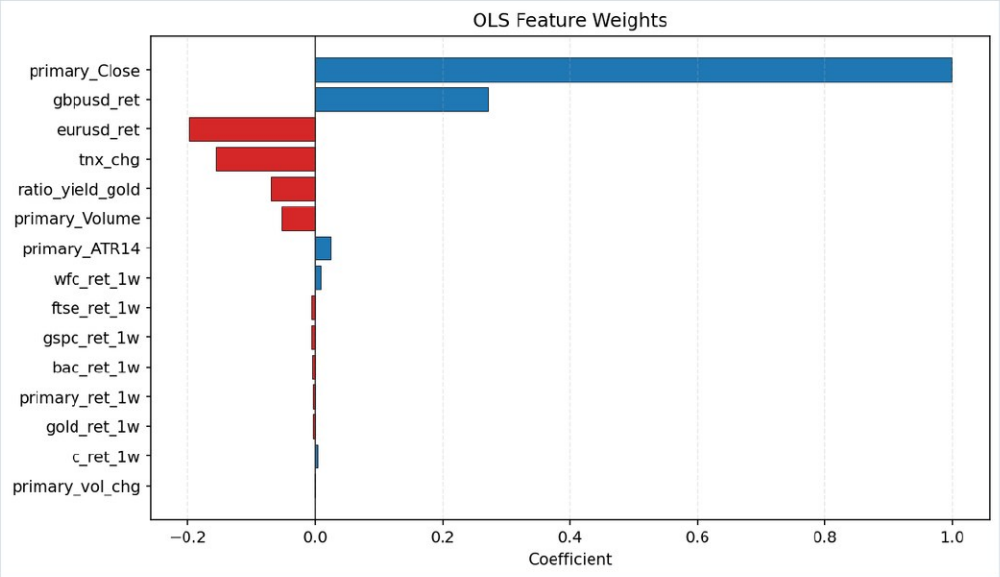
$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Higher R^2 = better model performance

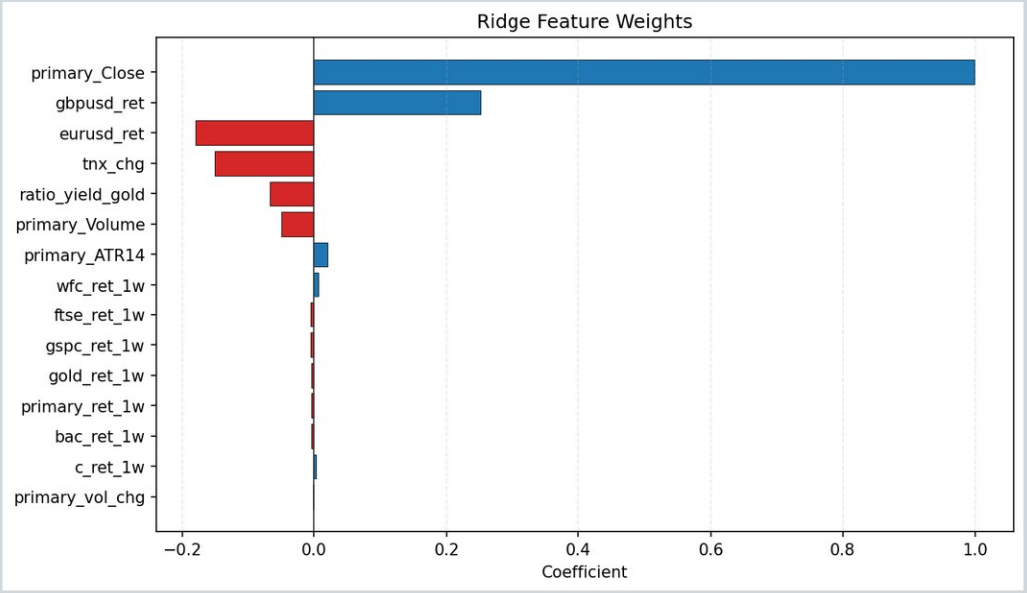
- **Lasso: R^2 = 0.994163** ★
- **XGBoost performed the worst: R^2 = 0.993802** ✕

Feature coefficients (OLS and Ridge)

OLS



Ridge

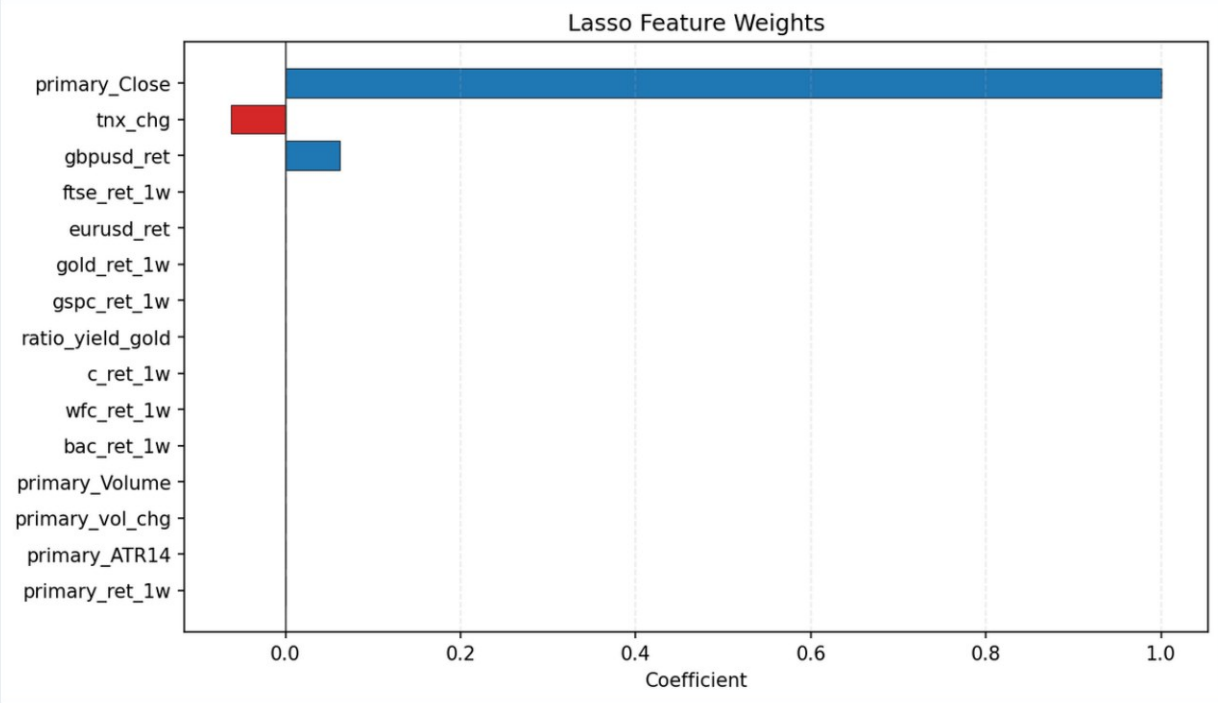


Feature coefficients (Lasso)

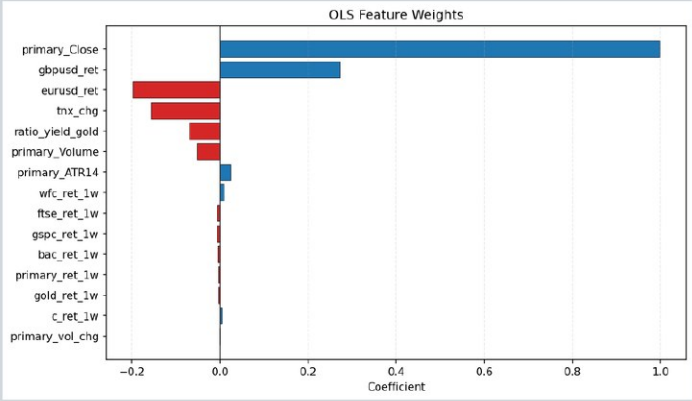
Lasso use **L1 regularization** based on OLS

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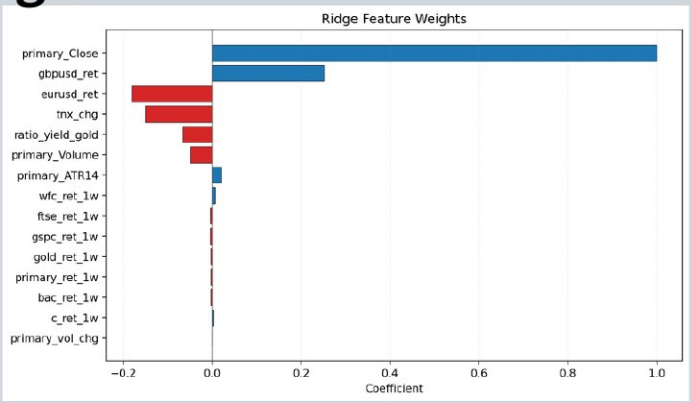
Lasso



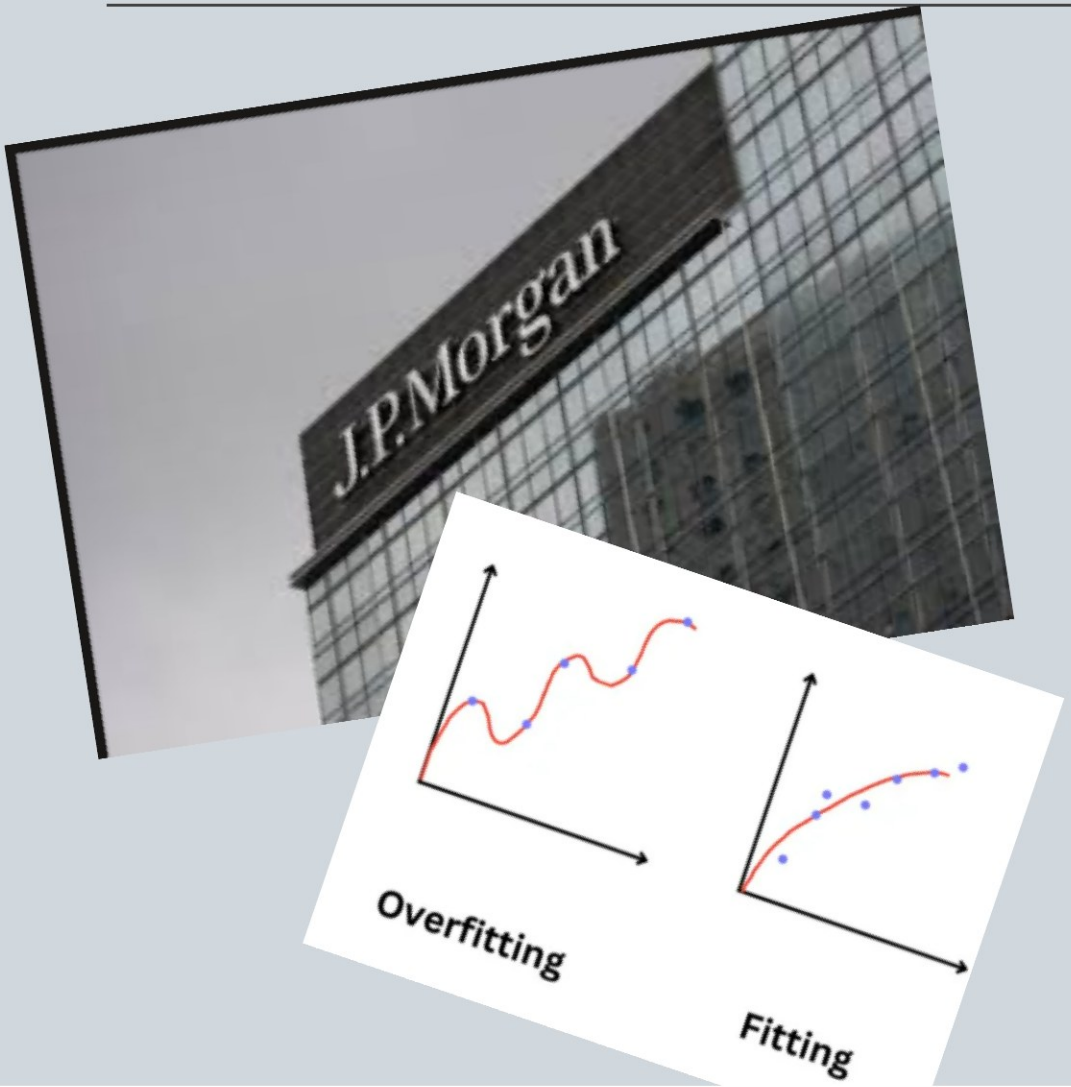
OLS



Ridge



Summarize



1. Completed the entire process from data acquisition, cleaning, feature engineering to model training and evaluation;
2. Verified the advantages of the Lasso regularization model in financial time series prediction;
3. Proved that a simple and efficient model (Lasso) is suitable for short-term stock price prediction. (Lasso > XGBoost)

Thank You